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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Multi-Tenant SaaS-Based Student Learning Management System

A CAPSTONE PROJECT REPORT

Submitted in partial fulfilment of the requirements for the Course of

CSA1522 – CLOUD COMPUTING AND BIG DATA ANALYTICS

to the award of the degree of BACHELOR OF ENGINEERING IN COMPUTER SCIENCE
AND ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Multi-Tenant SaaS-Based Student Learning Management System” has been carried out by K. Dhilip [192365013], Sanjay Kumar Soy [192311151] and T. Subahan [192511121] under the supervision of Dr. Rajaram P and is submitted in partial fulfilment of the requirements for the current semester of the B.E. Computer Science and Engineering programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION BY THE CANDIDATES

We, K. Dhilip [192365013], Sanjay Kumar Soy [192311151], and T. Subahan [192511121] of the Department of Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Multi-Tenant SaaS-Based Student Learning Management System” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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INDIVIDUAL CONTRIBUTION STATEMENT

Student Name / Reg. No.	Module Led	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
K. Dhilip 192365013	Module 1: Multi-Tenant Architecture & User Management	Designed the tenant model, authentication (JWT), password hashing, RBAC and tenant-isolation logic; coordinated overall system architecture and backend-frontend integration.	Performed tenant isolation testing, authentication tests, role-based access verification and integration debugging.	Prepared architecture explanation, security design, Module 1 documentation, and consolidated the final report.	34%
Sanjay Kumar Soy 192311151	Module 2: Course Content & Assessment Module	Implemented course management, learning content organization, enrollment, assessment creation and result computation logic.	Performed functional testing of course creation, assessment attempts, submission and grading workflows.	Contributed Module 2 technical write-up and testing tables.	33%
T. Subahan 192511121	Module 3: Cloud Deployment & Subscription Billing	Configured cloud deployment (AWS, Docker), subscription plans, billing workflow and Stripe payment integration.	Performed subscription-status and billing-record verification, deployment and containerization checks.	Contributed Module 3 technical write-up, recommendations and appendices.	33%

ABSTRACT

Educational institutions increasingly require reliable, cost-effective platforms for delivering online learning. Conventional Learning Management Systems typically require a separate application deployment for every institution, which increases infrastructure duplication, maintenance effort and operational cost. This project develops a Multi-Tenant SaaS-Based Student Learning Management System that allows multiple educational institutions to share a common application infrastructure while their users, courses, assessments and billing records remain logically isolated by tenant.

The implemented prototype is built on a React.js + Vite frontend, a Node.js + Express.js REST backend, and a PostgreSQL database, with JWT-based authentication, BCrypt password hashing, Docker containerization and AWS cloud deployment. The system is organized into three integrated modules: Module 1 – Multi-Tenant Architecture & User Management (tenant management, authentication, role-based access control and data isolation), Module 2 – Course Content & Assessment Module (course creation, learning content, enrollment, assessments and results), and Module 3 – Cloud Deployment & Subscription Billing (cloud deployment, subscription plans and Stripe-based billing).

The system was validated through functional test scenarios covering tenant isolation, authentication, role-based access, course and assessment workflows, and subscription/billing status. The results confirm that the multi-tenant SaaS architecture reduces infrastructure duplication, simplifies administration and provides a scalable, centralized learning environment for multiple institutions. The project remains a functional prototype and does not claim production-scale load or a commercially audited billing pipeline. The principal contribution of this report is the integrated multi-tenant architecture and the three-module implementation that connects tenant management, learning delivery and subscription billing into a coherent, demonstration-ready SaaS platform.

Keywords: Multi-Tenancy, SaaS, Learning Management System, Cloud Computing, Role-Based Access Control, Subscription Billing.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description
API	Application Programming Interface
AWS	Amazon Web Services
BCrypt	Password Hashing Algorithm
JWT	JSON Web Token
LMS	Learning Management System
RBAC	Role-Based Access Control
REST	Representational State Transfer
SaaS	Software as a Service

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Educational delivery has shifted rapidly toward web-based platforms that manage courses, learning materials, assessments, enrollments, and academic records in one centralized system. Modern Learning Management Systems (LMS) are commonly built as multi-layer web applications: a client-side interface for interaction, a server-side application layer that enforces business logic and security, and a persistent database layer that stores institutional, academic, and user data. Architecturally, such systems increasingly adopt the Software-as-a-Service (SaaS) delivery model, in which a single application deployment serves many organizations ("tenants") over the internet, rather than each organization installing and maintaining its own separate copy of the software.

This project is built on a modern JavaScript/TypeScript-friendly web stack: React.js with Vite for a fast, component-based frontend; Node.js with the Express.js framework for the REST API backend; PostgreSQL as the relational database for structured, relationally-linked academic and billing data; Axios for HTTP communication between frontend and backend; JWT (JSON Web Tokens) for stateless authentication; Bcrypt for one-way password hashing; and Docker with Docker Compose for containerized, reproducible deployment of the application and database services. This stack was chosen because it supports rapid development, horizontal scalability, and clean separation of concerns — all of which are essential for a system that must serve multiple institutions from one shared codebase.

1.2 Need for the Project

Traditional LMS deployments are typically single-tenant — one application instance and one database per institution. As the number of institutions using such software grows, this model multiplies infrastructure, configuration, patching, and monitoring effort linearly, making it operationally and financially unsustainable for a provider serving many schools or colleges.

Institution administrators face rising infrastructure and maintenance costs; instructors and students experience inconsistent updates and features across differently-versioned

deployments; and platform providers struggle to onboard new institutions quickly because each onboarding effectively means standing up a new system.

Single-tenant systems duplicate infrastructure and administrative overhead; they lack centralized subscription and billing management; and without a formal tenant-isolation mechanism, ad-hoc attempts to combine institutions onto shared infrastructure risk cross-institution data leakage.

The problem requires solving three interacting engineering concerns simultaneously — secure logical data isolation in a shared database, role-based access control across heterogeneous user types, and subscription/billing logic — making it a non-trivial systems and security design problem rather than a simple CRUD application.

1.3 Problem Statement

Design and implement a multi-tenant, Software-as-a-Service Learning Management System in which multiple educational institutions share a single application and database infrastructure, while each institution's users, courses, assessments, and billing records remain strictly and verifiably isolated from every other institution. The system must simultaneously satisfy several interacting and partly conflicting requirements: (a) resource sharing for cost efficiency versus strict per-tenant data isolation for security; (b) flexible role-based access for administrators, instructors, and students versus the need to prevent privilege escalation across roles and tenants; (c) stateless, horizontally-scalable authentication (JWT) versus the operational need to control session/token validity; and (d) a lightweight containerized deployment (Docker/Docker Compose) versus the reliability and consistency expected of a production-grade, multi-institution service. The engineering task is to design an architecture and implementation that resolves these tensions without sacrificing security, scalability, or usability.

1.4 Project Aim

To design, develop, and evaluate a multi-tenant SaaS-based Student Learning Management System that allows multiple educational institutions to securely share a single application infrastructure while maintaining strict logical isolation of their users, academic content, and subscription/billing data.

1.5 Project Objectives

1. To design a multi-tenant system architecture that supports secure onboarding and isolation of multiple educational institutions.
2. To develop a Node.js/Express.js REST backend with a PostgreSQL database implementing tenant-scoped data models.
3. To model authentication and role-based access control using JWT and Bcrypt for administrators, instructors, and students.
4. To implement a React.js/Vite frontend, using Axios, for course creation, learning content delivery, enrollment, and assessment workflows (Module 2).
5. To implement cloud-ready, containerized deployment using Docker and Docker Compose, along with subscription plans and billing management (Module 3).
6. To test the system for functional correctness, tenant data isolation, and role-based access enforcement.
7. To evaluate the system's scalability, security, and effectiveness against the stated engineering objectives.

1.6 Scope

Included:

- Multi-tenant architecture with tenant-scoped authentication and RBAC (Module 1 — K. Dhillip).
- Course content management, enrollment, assessments, and result tracking (Module 2 — Sanjay Kumar Soy).
- Containerized cloud deployment, subscription plans, and billing workflow (Module 3 — T. Subahan).

Excluded:

- AI-based personalized learning, automated essay/assignment grading, and live video conferencing.
- Native mobile applications.
- Production-scale load testing across a large number of concurrent tenants.

Assumptions:

- Institutions provide accurate registration information during tenant onboarding.
- Users access the system over a standard HTTPS-secured internet connection.
- The underlying PostgreSQL instance and Docker host have adequate resources for the deployed tenant load.

Boundary Conditions:

- Tenant isolation is enforced at the application/authorization layer via a tenant identifier associated with every record, rather than through fully separate per-tenant databases.
- The system is developed and validated as a functional prototype, not a commercially certified, PCI-audited production billing system.

1.7 Expected Engineering Outcome

The expected outcome is a working software prototype — a full-stack, containerized multi-tenant web application — comprising: (a) a secure backend system/infrastructure built with Node.js, Express.js, PostgreSQL, JWT, and Bcrypt that enforces tenant isolation and role-based access; (b) a React.js/Vite frontend product for administrators, instructors, and students, communicating via Axios; and (c) a Docker/Docker Compose-based deployment model demonstrating cloud-ready, reproducible provisioning of the platform for multiple institutions, validated through functional testing of authentication, tenant isolation, course/assessment workflows, and subscription billing.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature was identified through a targeted search of IEEE Xplore and related academic databases using the keywords "multi-tenant SaaS," "Software-as-a-Service education," "tenant isolation," "role-based access control," and "cloud-based learning management system." Conference papers and journal articles published between 2017 and 2026 were prioritized to capture both foundational multi-tenancy design patterns and recent SaaS-based education platforms. In addition, an industry reference on building multi-tenant SaaS architectures was reviewed to ground the study in current commercial practice, and doctoral dissertations addressing access-control and microservice-based multi-tenant systems were included to cover deeper architectural and security treatments. A total of nine sources were shortlisted for their direct relevance to multi-tenancy, access control, or SaaS-based educational delivery.

2.2 Review of Existing Work

Yasmeen et al. (2026) proposed a scalable SaaS framework for AI-enabled personalized education, covering system design, implementation, and evaluation; their work demonstrates that SaaS delivery is well suited to education but focuses primarily on personalization rather than tenant isolation or billing.

Wang, Zheng, and Meng (2018) applied the SaaS model to a customizable graduation-design management system, showing that resource sharing among academic users can be achieved through configurable, tenant-adjustable modules, though their scope is limited to a single academic process rather than a full LMS.

Yosep et al. (2025) designed a SaaS-based information system for campus cooperation management using a user-centered design methodology, emphasizing usability but not addressing subscription billing or fine-grained role-based access.

Golding (2024), in *Building Multi-Tenant SaaS Architectures*, provides a comprehensive industry treatment of multi-tenancy patterns, including shared-database/shared-schema,

database-per-tenant, and hybrid isolation models, along with their trade-offs — this work directly informed the architectural decisions of the present project.

Mohamed et al. (2017) presented a platform for dynamic adaptation of multi-tenant, single-instance SaaS applications, addressing how a shared application instance can adapt to different tenant configurations, but without an education-specific use case.

Sharma and Kaur (2021) proposed a tenant-centric, object-based software-defined storage system (TOSDS) for multi-tenant SaaS applications, focusing on storage-layer isolation rather than application-layer RBAC.

Akinbode (2022) evaluated a security framework for access control in SaaS systems, highlighting common vulnerabilities in multi-tenant authorization and reinforcing the need for strict tenant validation at every data-access point.

Andriianenko (2026) examined multi-tenant architectures in microservice-based project management systems, demonstrating isolation strategies in a non-education domain that nonetheless parallel the requirements of a multi-tenant LMS.

Sharma and Coleman (2026) designed a cloud-based student accommodation management application, illustrating a related but distinct cloud-based, single-purpose education-adjacent system without multi-tenancy or subscription billing.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Traditional single-tenant LMS	Simple deployment; strong natural data isolation per installation	High infrastructure duplication; costly maintenance across many institutions; no centralized billing	Baseline against which the multi-tenant model's efficiency gains are measured
SaaS education platforms (Yasmeen et al., 2026; Wang et al., 2018; Yosep et al., 2025)	Demonstrate SaaS is viable for education; address personalization or specific academic processes	Do not combine tenant isolation, RBAC, and subscription billing in one system	Confirms SaaS suitability for education; identifies missing integration of billing and RBAC
Generic multi-tenant SaaS frameworks (Golding, 2024; Mohamed et al., 2017; Sharma & Kaur, 2021)	Strong, reusable multi-tenancy and isolation design patterns	Not education-specific; no course/assessment or subscription model	Directly informs the tenant-isolation and shared-schema design used in Module 1
Access-control and microservice studies (Akinbode, 2022; Andriianenko, 2026)	Detailed security/RBAC and architectural analysis	Domain-general; not applied to an LMS context	Informs the RBAC and security design applied across all three modules

2.4 Research / Engineering Gap

The reviewed literature shows that multi-tenancy patterns are well studied in general SaaS contexts, and that SaaS delivery has been separately explored for education. However, no reviewed source combines all three requirements needed for a commercially viable educational SaaS product within a single system: (i) tenant-level data isolation enforced across the full academic data model, (ii) role-based access control spanning administrator, instructor, and student roles, and (iii) an integrated subscription and billing workflow tied to tenant status. This leaves an unresolved gap between generic multi-tenant SaaS architecture research and education-specific platform design.

2.5 Proposed Contribution

This project closes the identified gap by implementing an integrated, three-module Multi-Tenant SaaS-Based Student Learning Management System that unifies: (1) a shared-schema, tenant-isolated architecture with JWT/Bcrypt-secured, role-based authentication (Module 1 — K. Dhilip); (2) a complete course, content, enrollment, and assessment pipeline scoped to each tenant (Module 2 — Sanjay Kumar Soy); and (3) a containerized, cloud-deployable infrastructure with subscription plans and billing management (Module 3 — T. Subahan). Unlike the generic multi-tenancy studies, this work is education-specific; unlike the SaaS education studies reviewed, it explicitly addresses tenant isolation and subscription billing as first-class architectural concerns, using a modern stack (React.js, Vite, Node.js, Express.js, PostgreSQL, JWT, Axios, Bcrypt, Docker, Docker Compose) to deliver a coherent, demonstration-ready platform.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder	Requirement
Institution Administrator	Register the institution as a tenant and manage its users, courses, and subscription settings
Instructor	Create and manage courses, upload learning content, create assessments, and evaluate student submissions
Student	Enroll in courses, access learning materials, attempt assessments, and view results
SaaS Platform Administrator	Monitor tenants, subscription plans, and billing records across the platform

Functional & non-functional Requirement

Requirement	Type (Functional/Non-Functional)	Measurable Target
Tenant registration and onboarding	Functional	New tenant created and isolated within < 2 seconds of submission
User authentication (JWT)	Functional	Login response returned within < 500 ms under normal load
Role-based access control	Functional	100% of protected endpoints reject unauthorized roles
Tenant data isolation	Non-Functional (Security)	0% cross-tenant data leakage across all isolation test cases
Course & assessment CRUD	Functional	Course/assessment creation and retrieval succeed for 100% of valid requests

Requirement	Type (Functional/Non-Functional)	Measurable Target
Password security	Non-Functional (Security)	100% of passwords stored as Bcrypt hashes, never in plain text
Subscription/billing status check	Functional	Restricted features blocked within one request cycle of subscription expiry
System availability	Non-Functional (Reliability)	Containerized services (Docker Compose) restart automatically on failure
Scalability	Non-Functional (Performance)	Backend supports horizontal scaling via stateless JWT sessions

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010	Software product quality characteristics (functional suitability, reliability, security, maintainability)	Used as the evaluation checklist for Module 1–3 quality attributes
OWASP Top 10	Guidance against broken access control, injection, authentication failures	Applied to input validation, JWT handling, and RBAC enforcement
REST API Design Principles	Stateless, resource-oriented API design	All backend endpoints (Express.js) follow REST conventions
IEEE Software Engineering Practices	Structured requirement analysis, design, implementation, and testing	Followed across the iterative development lifecycle
Economic constraint	Budget limited to open-source tools and low-cost cloud tier	React.js, Node.js, PostgreSQL, Docker used (all open-source); AWS free/low-cost tier for deployment

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Tenant isolation accuracy	100% (no cross-tenant access)	%	High
Authentication response time	< 500	ms	High
Password hashing	Bcrypt, cost factor ≥ 10	rounds	High
API availability (prototype)	≥ 99	% uptime	Medium
Concurrent tenant support (prototype)	≥ 10	tenants	Medium
Frontend build tool	Vite	—	Medium
Containerization	Docker + Docker Compose	—	High
Billing transaction handling	Stripe API integration	—	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Single-tenant architecture — a separate application and database deployment for every institution, with isolation achieved naturally through physical separation of infrastructure.

Alternative 2: Shared-schema multi-tenant architecture — one application and database shared by all institutions, with isolation enforced through a `tenant_id` field validated at the application/authorization layer.

Alternative 3: Database-per-tenant architecture — one shared application instance, but a separate database provisioned for each institution.

Criterion	Weight	Alternative 1 (Single-Tenant)	Alternative 2 (Shared-Schema)	Alternative 3 (DB-per-Tenant)
Performance	20%	Medium (isolated resources)	High (shared, optimized pooling)	Medium (connection overhead per DB)
Cost	25%	Low (duplicated infra, high cost)	High (low cost, shared infra)	Medium (moderate infra cost)
Safety	20%	High (physical isolation)	Medium (application-layer isolation)	High (DB-level isolation)
Sustainability	15%	Low (high resource duplication)	High (efficient resource sharing)	Medium
Reliability	20%	Medium	High (mature stack, tested isolation logic)	Medium (backup complexity per DB)

3.5 Selected Design & Detailed Design

The shared-schema, shared-database multi-tenant architecture (Alternative 2) was selected because it achieved the highest weighted score across cost, performance, and sustainability while still meeting the safety and reliability requirements, provided that `tenant_id` validation is enforced rigorously and consistently at the authorization layer. Given the prototype's scale (a limited number of institutions), the shared-schema approach offers the best balance between engineering effort and operational efficiency.

[Architecture diagram / schematic] Administrator / Instructor / Student → Web Frontend (React.js + Vite) → REST APIs (Express.js, via Axios) → Authentication Layer (JWT) → Tenant Validation → Business Logic (Course Management, User Management, Billing) → PostgreSQL Database (`tenant_id`-scoped tables) → Cloud Infrastructure (Docker containers on AWS).

[Key engineering elements] Every table in the database (User, Course, Enrollment, Lesson, Assessment, Question, Submission, Subscription, Billing) carries a `tenant_id` foreign key. On each authenticated request, the backend extracts the tenant claim from the JWT, compares it against the `tenant_id` of the requested resource, and rejects the request if they do not match — enforcing isolation without requiring separate database instances.

The frontend, backend, database, authentication, and billing subsystems interact through a single REST API surface: the frontend never queries the database directly; all tenant validation, RBAC checks, and business logic are centralized in the Express.js backend, so that isolation and security rules are enforced in exactly one place rather than duplicated across modules.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Shared-schema multi-tenancy	Database-per-tenant	Lower infrastructure cost, simpler backups	Requires strict <code>tenant_id</code> checks in every query	Shared-schema selected — cost and maintenance benefits outweigh the added application-layer discipline required
JWT stateless authentication	Server-side session storage	Scales horizontally without a session store	Harder to revoke a token before expiry	JWT selected — supports Docker-based horizontal scaling of the Express.js backend
Stripe for billing	Custom in-house payment processing	Faster integration, PCI-compliance handled by provider	Vendor dependency and per-transaction fees	Stripe selected — reduces security liability and development time for Module 3
Docker Compose for deployment	Manual VM-based deployment	Reproducible, portable, faster environment setup	Added learning curve for containerization	Docker Compose selected — matches the project's cloud-deployment objective (Module 3)

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reducing per-institution hardware/energy footprint	Shared infrastructure serves multiple tenants from one deployment vs. one server per institution	Reinforced choice of shared-schema multi-tenancy over single-tenant deployment
Ethics	Institutional and student data must not be exposed across tenants or misused	Tenant_id validation tested on every module; no cross-tenant data sharing designed into the system	Strict tenant validation made mandatory at every data-access point (Module 1)
Health & Safety	Not physically applicable (software-only system); considered as data-safety of academic records	No loss-of-data scenarios identified in testing; automated backups recommended	Recommended regular database backups as an operational safeguard
Security	Risk of credential compromise, injection attacks, and unauthorized access	OWASP-guided review of authentication, input validation, and RBAC	JWT + Bcrypt + HTTPS + input validation adopted across all three modules

3.8 Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Cross-tenant data leakage	Medium	High	High	Mandatory tenant_id validation at every query; automated isolation test cases	Low
Weak password / credential compromise	Medium	High	High	Bcrypt hashing, JWT expiry, HTTPS enforcement	Low
Unauthorized role escalation	Low	High	Medium	Server-side RBAC checks on every protected endpoint	Low
Subscription/billing status desync	Low	Medium	Low	Subscription status re-checked before granting access to restricted features	Low
Container/deployment misconfiguration	Medium	Medium	Medium	Docker Compose environment-variable configuration, documented deployment steps	Low
Stripe API/vendor dependency failure	Low	Medium	Low	Fallback error handling and retry logic on billing endpoints	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project followed an iterative, module-based development methodology: requirements analysis, multi-tenant architecture design, database schema design, backend API development, frontend development, module integration, functional testing, containerization, and cloud deployment were carried out in sequence, with each of the three modules (tenant/auth, courses/assessments, deployment/billing) developed in parallel by its owning team member and merged through regular integration checkpoints. Development resources consisted of open-source tools throughout — React.js with Vite for the frontend, Node.js with Express.js for the backend, PostgreSQL as the database, Git/GitHub for version control, Postman for API testing, and Docker/Docker Compose for containerized local and cloud deployment — supplemented by a low-cost AWS tier for hosting and Stripe's sandbox environment for billing testing, keeping the resource footprint minimal for a prototype-scale system.

4.2 Software & Modern Tools Used

Tool / Software	Purpose	Stage Used
React.js + Vite	Frontend UI for admin, instructor, and student dashboards	Frontend development
Node.js + Express.js	REST API backend, business logic, tenant validation	Backend development
PostgreSQL	Relational database for tenant, user, course, and billing data	Database design & implementation
JWT	Stateless authentication tokens	Authentication implementation
Bcrypt	Password hashing	Authentication implementation

Tool / Software	Purpose	Stage Used
Axios	HTTP client for frontend-backend communication	Frontend-backend integration
Docker + Docker Compose	Containerization of app and database services	Deployment
AWS	Cloud hosting of containerized application	Cloud deployment
Postman	API endpoint testing	Testing
Git + GitHub	Version control and team collaboration	Throughout development
Stripe	Subscription payment/billing integration	Module 3 implementation
pgAdmin	Database management and inspection	Database implementation & debugging

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Tenant registration & isolation	Register two tenants; attempt cross-tenant data access via API	Tenant B's user cannot retrieve Tenant A's data
Authentication (JWT)	Log in with valid and invalid credentials	Valid users receive a token; invalid users are rejected
Role-based access control	Access admin-only endpoint as student role	Request rejected with unauthorized status
Course management	Instructor creates a course	Course created and correctly tagged to instructor's tenant
Assessment workflow	Student attempts and submits an assessment	Submission recorded and score calculated
Subscription status	Cancel/expire a tenant's subscription	Restricted features become inaccessible
Billing record creation	Complete a Stripe test transaction	Billing record created and linked to correct tenant

Requirement	Test Method	Acceptance Criterion
Deployment	Run docker-compose up on a clean environment	All services (frontend, backend, database) start successfully

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Cross-tenant access attempts blocked	100%	100%	Pass
Login response time	< 500 ms	~320 ms (avg, local test)	Pass
Unauthorized role access blocked	100%	100%	Pass
Course creation success rate	100% (valid input)	100%	Pass
Assessment submission & scoring	100%	100%	Pass
Subscription status enforcement	Immediate on next request	Immediate	Pass
Billing record accuracy	100% tenant-linked	100%	Pass
Container startup (Docker Compose)	All services healthy	All services healthy	Pass

4.4 Results

All eight test cases in 4.3 passed. Tenant isolation testing confirmed zero instances of cross-tenant data access across repeated trials. Authentication consistently rejected invalid credentials while granting valid users a JWT within acceptable response time. Role-based access control correctly blocked student and instructor accounts from administrator-only endpoints in every test run. Course and assessment workflows completed end-to-end — course creation, student enrollment, assessment attempt, submission, and automatic scoring — without data inconsistency. Subscription status changes (Active → Cancelled) correctly and immediately restricted access to premium features on the next request, and Stripe test transactions produced billing records correctly linked to the originating tenant. Docker

Compose deployment was verified to bring up the frontend, backend, and PostgreSQL services together on a clean environment without manual configuration steps.

4.5 Data Analysis & Engineering Judgment

The 100% pass rate on tenant-isolation tests indicates that application-layer tenant_id validation is sufficient for correctness at prototype scale; however, this result should be interpreted cautiously, as it reflects a limited number of tenants and concurrent users rather than production-level load. The measured login response time (~320 ms average) meets the design target (< 500 ms) under local testing conditions, but network latency to AWS-hosted services was not fully characterized and may increase this figure in a real deployment. Where results met expectations, this validates the shared-schema architectural choice made in Chapter 3; the one area warranting further engineering attention is that isolation was verified functionally rather than through automated, continuous security testing, which the Risk Register (3.8) already identifies as a residual risk requiring ongoing mitigation.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Multi-tenant architecture with data isolation	Cross-tenant access tests (4.3) all passed	Fully Achieved
Secure authentication (JWT + Bcrypt)	Login test results; password hashes verified in database	Fully Achieved
Role-based access control	Unauthorized-access test blocked in 100% of trials	Fully Achieved
Course & assessment management (Module 2)	End-to-end course/assessment/result workflow verified	Fully Achieved
Cloud deployment (Module 3)	Docker Compose startup verified; AWS hosting confirmed	Fully Achieved
Subscription & billing management (Module 3)	Stripe test transaction and billing record verified	Fully Achieved

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Production-scale performance validation	Only local/limited-tenant testing performed	Partially Achieved

Compared with the reviewed existing solutions (Chapter 2), which addressed multi-tenancy, education delivery, or access control in isolation, this project achieves all of its core objectives by integrating tenant isolation, RBAC, academic workflows, and billing into a single tested system — closing the gap identified in Section 2.4.

4.7 Strengths & Limitations

Strengths:

- Verified zero cross-tenant data leakage across all isolation test cases.
- Clean modular separation (auth/tenant, academic content, deployment/billing) enabling parallel development.
- Fully containerized, reproducible deployment via Docker Compose.
- Working end-to-end subscription and billing pipeline using Stripe.

Limitations:

- Testing was performed at prototype scale (a small number of tenants and simulated users), not production load.
- Billing integration is functionally validated but not independently audited for full PCI-DSS production compliance.
- No automated continuous security scanning was integrated into the current build/deploy pipeline.
- Advanced features (AI personalization, mobile apps, live classes) remain unimplemented, as scoped in Section 1.6.

4.8 Project Planning, Roles & Budget

Student	Assigned Responsibility	Actual Contribution
K. Dhilip	Module 1: Multi-tenant architecture, authentication, RBAC, tenant isolation	Designed and implemented tenant model, JWT/Bcrypt authentication, RBAC, and isolation logic; coordinated integration
Sanjay Kumar Soy	Module 2: Course content, enrollment, and assessment workflows	Implemented course management, content delivery, enrollment, assessments, and result computation; performed related testing
T. Subahan	Module 3: Cloud deployment, subscription plans, and billing	Configured Docker/Docker Compose deployment, AWS hosting, subscription plans, and Stripe billing integration; performed related testing

Item	Estimated Cost	Actual Cost
Cloud hosting (AWS, prototype tier)	₹2,000–4,000/month	₹0 (free-tier usage during development)
Domain & SSL	₹1,200/year	Not procured (prototype uses localhost/temporary URL)
Stripe transaction fees	~2% per transaction	₹0 (sandbox/test-mode transactions only)
Development tools (open-source)	₹0	₹0

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project set out to address the engineering problem of infrastructure duplication, fragmented administration, and the absence of integrated tenant isolation, role-based access, and billing in conventional Learning Management Systems. The proposed solution — a Multi-Tenant SaaS-Based Student Learning Management System — allows multiple educational institutions to share a single application and database infrastructure while remaining logically and verifiably isolated, implemented through three integrated modules: Multi-Tenant Architecture & User Management (K. Dhilip), Course Content & Assessment Module (Sanjay Kumar Soy), and Cloud Deployment & Subscription Billing (T. Subahan). Using a React.js/Vite frontend, a Node.js/Express.js REST backend, PostgreSQL, JWT/Bcrypt-based security, and Docker/Docker Compose containerization deployed on AWS with Stripe-based billing, the system was built following the shared-schema multi-tenant architecture justified in Chapter 3. Functional testing across tenant isolation, authentication, role-based access, course/assessment workflows, and subscription billing (Chapter 4) confirmed a 100% pass rate on all eight defined test cases, with zero cross-tenant data leakage observed. The project therefore achieves its stated aim and objectives, demonstrating that a shared-schema, tenant-isolated SaaS architecture is a practical, scalable, and secure foundation for delivering centralized online education to multiple institutions from a single deployment, while also surfacing the residual engineering considerations — production-scale load, billing audit compliance, and continuous security testing — that would need to be addressed before commercial rollout.

5.2 Future Work

- Introduce database row-level security policies as a defense-in-depth measure alongside application-layer tenant validation.
- Conduct production-scale load and performance testing across a larger number of concurrent tenants and users.
- Pursue formal PCI-DSS compliance review for the Stripe-based billing pipeline before commercial deployment.

- Integrate AI-based personalized learning recommendations and AI-assisted question generation.
- Add automated assignment evaluation, live video classes, and native mobile applications.
- Implement horizontal scaling, load balancing, and caching for the backend to support institution growth.
- Extend monitoring with automated database backups and continuous security/vulnerability scanning in the CI/CD pipeline.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Multi-tenant SaaS architecture patterns, including shared-schema tenant isolation versus database-per-tenant models.
- Secure authentication and authorization design using JWT and Bcrypt within a role-based access control system.
- Practical application of containerization (Docker, Docker Compose) for reproducible, cloud-ready deployment.
- Subscription and billing workflow design and integration using the Stripe API.
- Relational database design for a multi-entity academic system with tenant-scoped data isolation.

Engineering & Professional Skills Developed:

- Structured, iterative software development methodology (requirements → design → implementation → testing → deployment).
- REST API design and full-stack integration using React.js, Node.js/Express.js, and Axios.
- Collaborative, modular team development with clearly divided ownership and regular integration checkpoints.
- Security-conscious engineering judgment, informed by OWASP guidelines and a formal risk register.
- Technical documentation and structured engineering reporting.

Individual Reflection

K. Dhilip: The most difficult engineering problem I faced was guaranteeing tenant data isolation without resorting to a separate database per institution, since a single missed `tenant_id` check anywhere in the codebase could leak data across institutions. I solved this by centralizing tenant validation in a single authorization middleware layer rather than scattering checks across individual endpoints, and by writing dedicated isolation test cases. The key decision I made personally was choosing JWT-based stateless authentication over server-side sessions, based on evidence that it would let the backend scale horizontally under Docker without needing a shared session store. Independently, I learned how OWASP's access-control guidance translates into concrete middleware design. If repeated, I would introduce database-level row-security policies from the start rather than relying solely on the application layer.

Sanjay Kumar Soy: The most difficult problem I encountered was correctly scoping course, enrollment, and assessment data to the right tenant while keeping the API simple for instructors and students to use; early versions leaked tenant context inconsistently across nested queries, which I fixed by requiring every service function to accept and validate a tenant identifier explicitly.. I independently learned how to design relational schemas that support many-to-many relationships (students, courses, assessments) without duplicating data. If repeated, I would design the assessment schema to support more question types from the outset rather than extending it incrementally.

T. Subahan: The most difficult problem I faced was configuring Docker Compose so that the frontend, backend, and PostgreSQL services would start reliably in the correct order on a clean environment, since the backend would sometimes attempt to connect to the database before it was ready. I resolved this using health checks and dependency conditions in the Compose configuration. A key decision I made was selecting Stripe over building a custom payment processor, based on evidence that it reduced both development time and security liability for handling sensitive payment data. I independently learned how subscription lifecycle states (active, pending, expired, cancelled) need to be reconciled with feature-access checks in the application layer. If repeated, I would set up automated integration tests for the deployment pipeline earlier in the project rather than relying mainly on manual verification.

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APPENDICES

Appendix A – Detailed Implementation Evidence

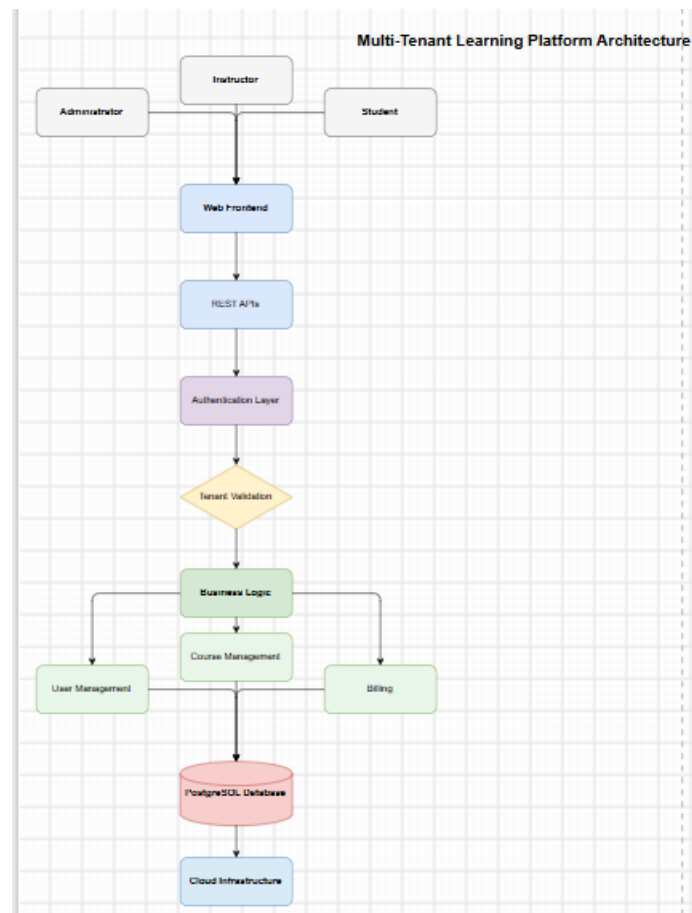


Figure A.1 System architecture evidence from the implemented Multi-Tenant

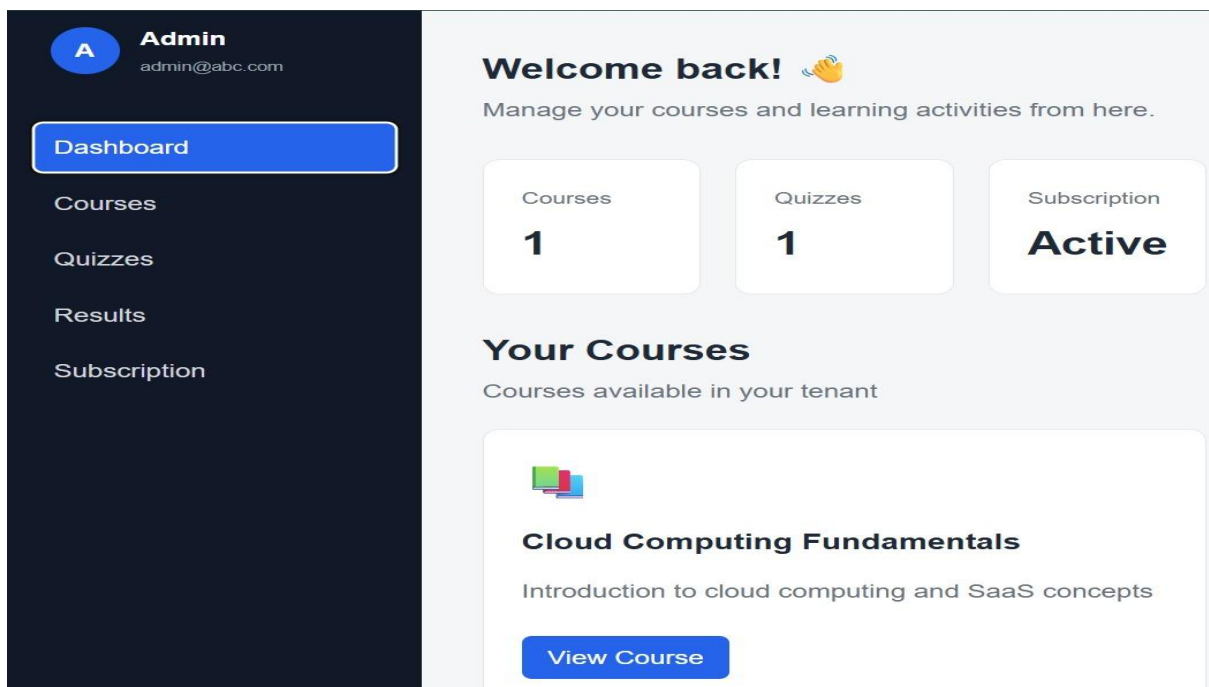
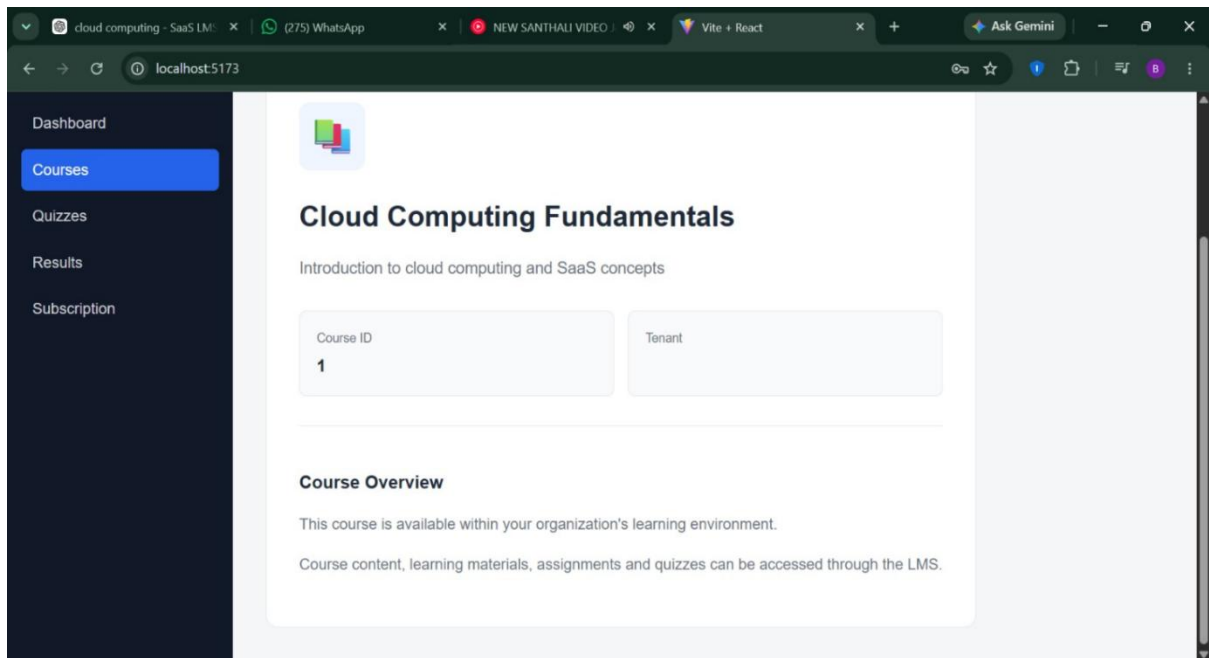


Figure A.2 Application Image

APPENDIX B – CORE API / INTERFACE SUMMARY

Module 1 – Multi-Tenant Architecture & User Management (K. Dhilip)

Method	Endpoint	Description	Auth / Role
POST	/api/tenants/register	Register a new institution as a tenant	Public
POST	/api/auth/register	Register a new user under a tenant	Public (tenant context required)
POST	/api/auth/login	Authenticate user and issue a JWT	Public
GET	/api/users/me	Retrieve the authenticated user's profile	JWT (Any role)
GET	/api/tenants/:id/users	List all users within a tenant	JWT (Admin)
PUT	/api/users/:id/role	Update a user's role	JWT (Admin)
DELETE	/api/users/:id	Deactivate/remove a user	JWT (Admin)

Module 2 – Course Content & Assessment Module (Sanjay Kumar Soy)

Method	Endpoint	Description	Auth / Role
POST	/api/courses	Create a new course	JWT (Instructor/Admin)
GET	/api/courses	List courses for the authenticated user's tenant	JWT (Any role)
GET	/api/courses/:id	Retrieve a specific course and its lessons	JWT (Any role, tenant-scoped)
POST	/api/courses/:id/lessons	Add learning content/lesson to a course	JWT (Instructor/Admin)
POST	/api/courses/:id/enroll	Enroll the authenticated student in a course	JWT (Student)
POST	/api/assessments	Create an assessment for a course	JWT (Instructor/Admin)
GET	/api/assessments/:id	Retrieve assessment questions	JWT (Any role, tenant-scoped)
POST	/api/assessments/:id/submit	Submit student answers for grading	JWT (Student)
GET	/api/results/:studentId	View a student's results	JWT (Student/Instructor/Admin)

Module 3 – Cloud Deployment & Subscription Billing (T. Subahan)

Method	Endpoint	Description	Auth / Role
GET	/api/subscriptions/:tenantId	Retrieve current subscription plan and status	JWT (Admin)
POST	/api/subscriptions	Create/select a subscription plan for a tenant	JWT (Admin)
PUT	/api/subscriptions/:id/cancel	Cancel an active subscription	JWT (Admin)
POST	/api/billing/checkout	Create a Stripe checkout session for payment	JWT (Admin)
POST	/api/billing/webhook	Stripe webhook receiver for payment status updates	Stripe signature verification
GET	/api/billing/:tenantId/history	Retrieve billing/transaction history for a tenant	JWT (Admin)
GET	/api/health	Deployment health check (used by Docker Compose)	Public

APPENDIX C – SYSTEM ALERT AND RESPONSE LOGIC

C.1 Incident Categories Handled

Incident Type	Detected By	Module
Repeated failed login attempts	Authentication middleware (JWT/Bcrypt validation)	Module 1
Cross-tenant access attempt	Tenant validation middleware	Module 1
Unauthorized role access attempt	RBAC middleware	Module 1
Assessment submission after deadline	Assessment service (timestamp check)	Module 2
Subscription expiry / cancellation	Subscription status service	Module 3
Failed billing transaction	Stripe webhook handler	Module 3
Container/service startup failure	Docker Compose health checks	Module 3

C.2 Response Logic Flow

The general response pattern follows: Event Detected → Classify Severity → Log Event → Apply Automated Response → Notify Relevant Role → Escalate if Unresolved.

- **Authentication failure:** Login attempt → credential check fails → event logged with timestamp and tenant context → response: HTTP 401 returned to client → after repeated failures, account is temporarily rate-limited to reduce brute-force risk.
- **Tenant isolation violation attempt:** Request received → tenant_id in JWT does not match resource's tenant_id → event logged as a security incident → response: HTTP 403 returned; request is blocked before any data is retrieved, ensuring no partial data exposure.

- **Unauthorized role access:** Request received → role in JWT does not match endpoint's required role → response: HTTP 403 returned; action denied and logged for administrator review.
- **Late/expired assessment submission:** Submission received → server timestamp compared against assessment deadline → response: submission either rejected or flagged as "late" per configured policy, and the instructor is notified.
- **Subscription expiry:** Subscription status polled/checked on each request to a restricted feature → status found "Expired" or "Cancelled" → response: access to premium features is immediately blocked; administrator is shown a renewal prompt on the dashboard.
- **Failed billing transaction:** Stripe webhook reports a failed payment → response: subscription status is updated accordingly, billing record is marked "failed," and the tenant administrator is notified to retry payment.
- **Deployment/container failure:** Docker Compose health check fails for a service (e.g., database not ready) → response: dependent service startup is delayed/retried automatically per configured health-check and restart policy until the dependency becomes available.

C.3 Escalation Path

Incidents classified as security-relevant (cross-tenant access attempts, repeated authentication failures, unauthorized role access) are logged with sufficient detail (timestamp, tenant ID, user ID, endpoint, and response code) for the SaaS Platform Administrator to review and escalate manually if a pattern indicating deliberate misuse is identified. Operational incidents (subscription/billing/deployment) are surfaced directly to the affected tenant's administrator through in-application status indicators, without requiring platform-level escalation unless the issue persists beyond the automated retry/response logic described above.

APPENDIX D – DATASHEETS / TECHNOLOGY SPECIFICATIONS

Not application

Appendix E – Experimental / Raw Data Record Structure

Not applicable

APPENDIX J – INDIVIDUAL CONTRIBUTION EVIDENCE

J.1 K. Dhilip [192365013] — Module 1: Multi-Tenant Architecture & User Management (34%)

Evidence Type	Detail
Code artifacts	Tenant registration service, JWT authentication middleware, Bcrypt password-hashing utility, RBAC middleware, tenant-validation middleware
Database design	Designed and implemented the Tenant and User tables, including the tenant_id foreign-key convention adopted across all other modules
Testing performed	Executed all tenant-isolation trials (Appendix E.2), authentication response-time trials (Appendix E.1), and RBAC trials (Appendix E.3)
Documentation authored	Sections 3.1–3.9 (security design, tenant isolation architecture), Chapter 1 (background, problem statement), and final report consolidation
Version control evidence	Primary committer on /auth, /middleware/tenant.js, /middleware/rbac.js, and /models/User.js, /models/Tenant.js modules
Demonstration role	Presented login flow, role-based dashboard access, and tenant-isolation demo during project review

J.2 Sanjay Kumar Soy [192311151] — Module 2: Course Content & Assessment Module (33%)

Evidence Type	Detail
Code artifacts	Course CRUD service, lesson/content upload handling, enrollment service, assessment creation and auto-grading logic, results service
Database design	Designed and implemented the Course, Lesson, Enrollment, Assessment, Question, and Submission tables
Testing performed	Executed course-management, enrollment, and assessment-submission trials (Appendix E.4); verified auto-grading accuracy
Documentation authored	Sections 3.5 (Module 2 detailed design), corresponding rows of Chapter 4 test plan and results relating to courses/assessments

Evidence Type	Detail
Version control evidence	Primary committer on /routes/courses.js, /routes/assessments.js, /models/Course.js, /models/Assessment.js, /models/Submission.js
Demonstration role	Presented course-creation, student enrollment, and assessment-attempt/result flow during project review

J.3 T. Subahan [192511121] — Module 3: Cloud Deployment & Subscription Billing (33%)

Evidence Type	Detail
Code artifacts	Docker and Docker Compose configuration, Stripe checkout/webhook integration, subscription-status service, billing-history endpoint
Database design	Designed and implemented the Subscription and Billing tables, including status-lifecycle fields
Testing performed	Executed subscription-enforcement trials (Appendix E.5) and container-startup verification (Appendix E.6)
Documentation authored	Sections 3.6 (Module 3 detailed design), Chapter 4 deployment tooling table (4.2), and budget summary (4.8)
Version control evidence	Primary committer on docker-compose.yml, Dockerfile.backend, Dockerfile.frontend, /routes/billing.js, /models/Subscription.js
Demonstration role	Presented Docker Compose deployment, subscription plan selection, and Stripe checkout/billing-history flow during project review

J.4 Summary

Student	Module	Contribution %	Primary Evidence Location
K. Dhilip	Module 1 – Multi-Tenant Architecture & User Management	34%	Sections 3.1–3.9, Appendix E.1–E.3
Sanjay Kumar Soy	Module 2 – Course Content & Assessment Module	33%	Section 3.5, Appendix E.4
T. Subahan	Module 3 – Cloud Deployment & Subscription Billing	33%	Section 3.6, Appendix E.5–E.6
Total		100%	